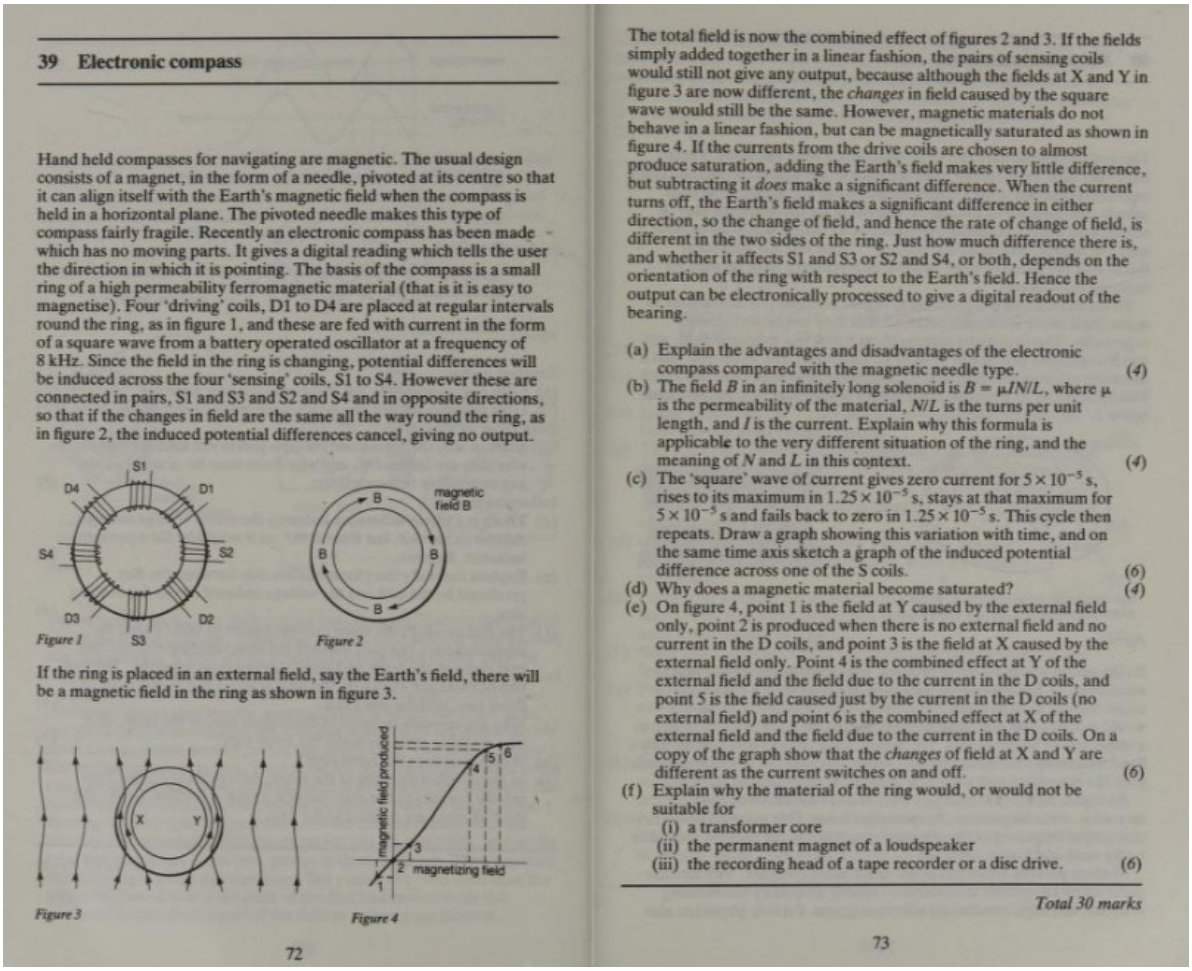


39 Electronic compass

Hand held compasses for navigating are magnetic. The usual design consists of a magnet, in the form of a needle, pivoted at its centre so that it can align itself with the Earth's magnetic field when the compass is held in a horizontal plane. The pivoted needle makes this type of compass fairly fragile. Recently an electronic compass has been made which has no moving parts. It gives a digital reading which tells the user the direction in which it is pointing. The basis of the compass is a small ring of a high permeability ferromagnetic material (that is it is easy to magnetise). Four 'driving' coils, D1 to D4 are placed at regular intervals round the ring, as in figure 1, and these are fed with current in the form of a square wave from a battery operated oscillator at a frequency of 8 kHz. Since the field in the ring is changing, potential differences will be induced across the four 'sensing' coils, S1 to S4. However these are connected in pairs, S1 and S3 and S2 and S4 and in opposite directions, so that if the changes in field are the same all the way round the ring, as in figure 2, the induced potential differences cancel, giving no output.

Figure 1 Figure 2

If the ring is placed in an external field, say the Earth's field, there will be a magnetic field in the ring as shown in figure 3.

Figure 3 Figure 4

72

The total field is now the combined effect of figures 2 and 3. If the fields simply added together in a linear fashion, the pairs of sensing coils would still not give any output, because although the fields at X and Y in figure 3 are now different, the *changes* in field caused by the square wave would still be the same. However, magnetic materials do not behave in a linear fashion, but can be magnetically saturated as shown in figure 4. If the currents from the drive coils are chosen to almost produce saturation, adding the Earth's field makes very little difference, but subtracting it *does* make a significant difference. When the current turns off, the Earth's field makes a significant difference in either direction, so the change of field, and hence the rate of change of field, is different in the two sides of the ring. Just how much difference there is, and whether it affects S1 and S3 or S2 and S4, or both, depends on the orientation of the ring with respect to the Earth's field. Hence the output can be electronically processed to give a digital readout of the bearing.

(a) Explain the advantages and disadvantages of the electronic compass compared with the magnetic needle type. (4)

(b) The field B in an infinitely long solenoid is $B = \mu IN/L$, where μ is the permeability of the material, N/L is the turns per unit length, and I is the current. Explain why this formula is applicable to the very different situation of the ring, and the meaning of N and L in this context. (4)

(c) The 'square' wave of current gives zero current for 5×10^{-5} s, rises to its maximum in 1.25×10^{-5} s, stays at that maximum for 5×10^{-5} s and fails back to zero in 1.25×10^{-5} s. This cycle then repeats. Draw a graph showing this variation with time, and on the same time axis sketch a graph of the induced potential difference across one of the S coils. (6)

(d) Why does a magnetic material become saturated? (4)

(e) On figure 4, point 1 is the field at Y caused by the external field only, point 2 is produced when there is no external field and no current in the D coils, and point 3 is the field at X caused by the external field only. Point 4 is the combined effect at Y of the external field and the field due to the current in the D coils, and point 5 is the field caused just by the current in the D coils (no external field) and point 6 is the combined effect at X of the external field and the field due to the current in the D coils. On a copy of the graph show that the *changes* of field at X and Y are different as the current switches on and off. (6)

(f) Explain why the material of the ring would, or would not be suitable for
(i) a transformer core
(ii) the permanent magnet of a loudspeaker
(iii) the recording head of a tape recorder or a disc drive. (6)

Total 30 marks

73

CIE 9702 Physics P4 Style Question – Electronic Compass (Passage-Based)

A company is developing an **electronic compass** for use in handheld navigation devices. The sensor head contains a soft magnetic ring with four equally spaced drive coils, D1 to D4, and four sensing coils, S1 to S4. An alternating current in the drive coils magnetises the ring so that the magnetic flux around the ring changes with time.

When there is **no external magnetic field**, the arrangement is symmetric and the induced e.m.f.s in opposite sensing coils are equal, so the net output is zero. When the ring is placed in the Earth's magnetic field, the total magnetic field in the ring becomes non-uniform and this symmetry is broken. As a result, a measurable output appears across a pair of sensing coils and can be processed to determine bearing.

To investigate this, a student drives the coils with a square-wave current of frequency $f = 8\,\text{kHz}$. For one half-cycle, the current rises from 0 to 1.25×10^{-5} A in 5.0×10^{-5} s, remains constant for 5.0×10^{-5} s, and then falls to zero in 1.25×10^{-5} s. This pattern repeats.

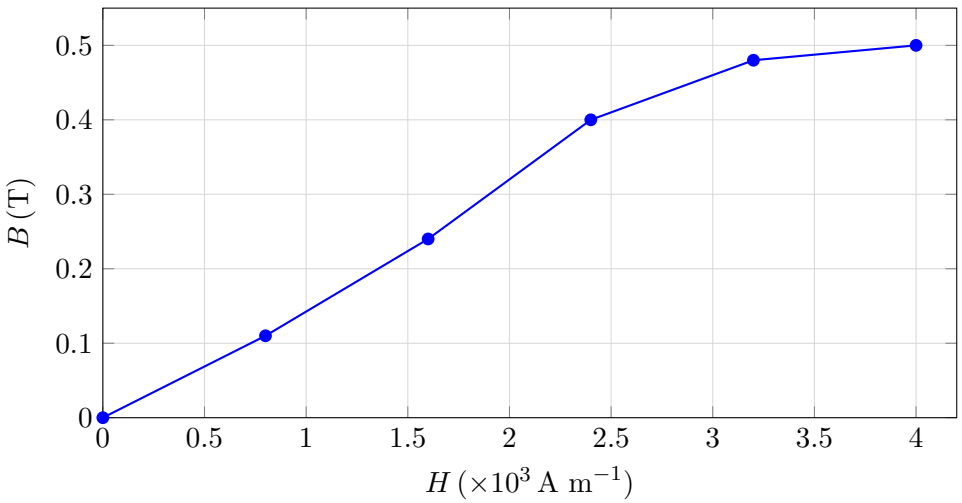
The student also measures the ring material's magnetisation response and obtains the following

data.

Table 1 shows magnetic flux density B as a function of magnetising field strength H.

H ($\times 10^3$ A m^{-1})	0.0	0.8	1.6	2.4	3.2	4.0
B (T)	0.00	0.11	0.24	0.40	0.48	0.50

Figure 1 is a graph of B against H for this material.



(a) Use the passage to explain why the net output of the sensing coils is approximately zero when there is no external field, and non-zero when the ring is in the Earth's field. [4]

(b) The magnetic flux density in a material is often written as

$$B = \mu n I / l,$$

where μ is permeability, n/l is turns per unit length, and I is current. State two assumptions behind this expression and explain why they are not fully valid for the ring sensor in this passage. [5]

(c)

(i) Sketch one complete cycle of current I against time for the drive coils, using the timing data in the passage. [3]

(ii) On the same time axis, sketch the induced potential difference across one sensing coil. Justify the sign and relative magnitude of the induced potential difference in each time interval. [5]

(d) Using Table 1 and Figure 1, determine the range of H over which the material begins to saturate. Explain why operating in this region can improve compass sensitivity to an external field. [5]

(e) A simplified model gives output voltage

$$V_{\text{out}} = k\left(\frac{\mathrm{d}\Phi_X}{\mathrm{d}t} - \frac{\mathrm{d}\Phi_Y}{\mathrm{d}t}\right),$$

where Φ_X and Φ_Y are fluxes at opposite points X and Y on the ring. Describe how turning the compass through 180° in the Earth's field changes V_{out}. State one limitation of this model. [4]

(f) Open-ended design question: Propose one modification to the sensor system to improve measurement reliability in real-world use (for example, nearby metal objects, temperature drift, or electrical noise). Your answer should include:

- a clear physical reason for the improvement,
- how you would test whether it works,
- one trade-off introduced by your design.

[4]

[**Total: 30 marks**]

Suggested Answers (Marking Guidance)

(a) No external field gives a symmetric flux distribution around the ring, so opposite sensing coils have equal induced e.m.f.s that cancel in the output pair. With Earth's field present, the total field differs at opposite points, making flux changes unequal and giving a non-zero differential output.

(b) Assumptions include a uniform field in a long solenoid and constant permeability. In the ring sensor, path geometry is curved and non-uniform, and permeability changes with operating point due to non-linear magnetisation and saturation.

(c) (i) Current waveform: linear rise for 5.0×10^{-5} s, flat for 5.0×10^{-5} s, linear fall for 1.25×10^{-5} s, then repeat. (ii) Since $\mathcal{E} = -N\,\mathrm{d}\Phi/\mathrm{d}t$, induced p.d. is approximately constant non-zero during rising and falling intervals, near zero on the flat section, and opposite polarity on rise vs fall.

(d) From about $H \approx 2.4\times10^3$ A m^{-1} onward, slope $\Delta B/\Delta H$ decreases strongly, indicating onset of saturation. In this region, adding/subtracting Earth's field changes local slope asymmetrically around the ring, increasing differential response between opposite sensing positions.

(e) A 180° rotation swaps which side experiences larger flux-change rate, so the bracketed term changes sign and V_{out} reverses polarity (similar magnitude ideally). Limitation: model ignores hysteresis, phase delay, and spatial non-uniformity of permeability.

(f) Example: add a second orthogonal sensing ring and perform vector combination plus digital filtering. This improves rejection of local directional interference and noise, then calibration can be tested by rotating the device in known fields and comparing bearing error before/after. Trade-off: higher power consumption and increased signal-processing complexity.

Variant 2 – Extended Data, Higher Difficulty

A research group is testing an upgraded electronic compass for drone navigation at high latitude, where magnetic inclination and environmental interference are significant. The ring sensor design is unchanged, but the drive electronics now allow selectable frequencies and amplitudes. The objective is to maximise directional sensitivity while maintaining linearity and low thermal drift.

The drive current remains piecewise linear in each cycle, but for this variant the timing is:

- rise from 0 to 2.00×10^{-5} A in 4.0×10^{-5} s,
- hold at maximum for 3.0×10^{-5} s,
- fall to zero in 1.0×10^{-5} s,
- remain at zero for 2.0×10^{-5} s,

and then repeat.

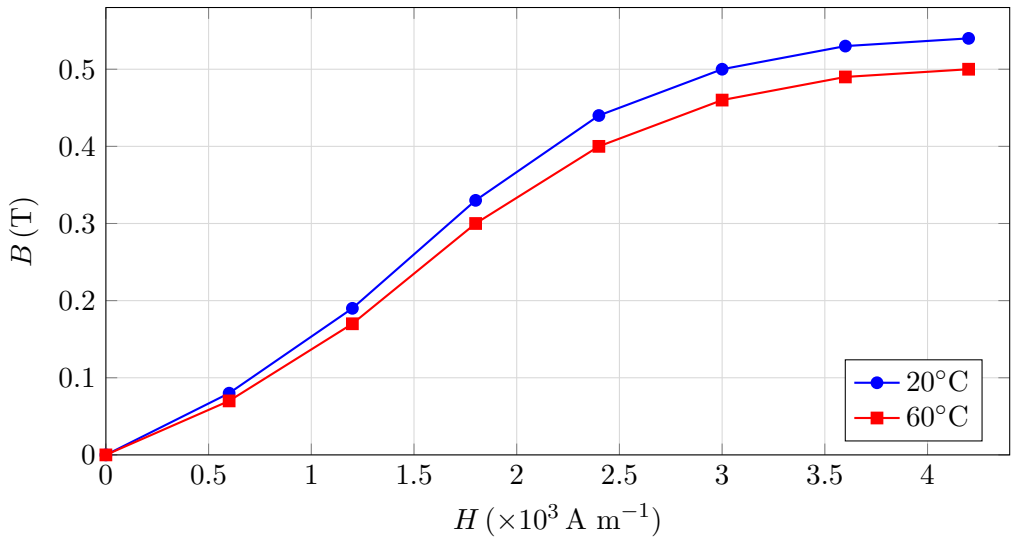
Table 2 gives magnetisation data for the ring material at two temperatures.

H ($\times 10^3$ A m^{-1})	0.0	0.6	1.2	1.8	2.4	3.0	3.6	4.2
B (T) at 20°C	0.00	0.08	0.19	0.33	0.44	0.50	0.53	0.54
B (T) at 60°C	0.00	0.07	0.17	0.30	0.40	0.46	0.49	0.50

Table 3 gives measured differential output amplitude $V_{\mathrm{out,pk}}$ for different external field strengths B_{ext}.

B_{ext} (μT)	0	10	20	30	40	50
$V_{\mathrm{out,pk}}$ (mV)	0.00	1.52	3.01	4.44	5.68	6.50

Figure 2 shows temperature-dependent B–H curves.



(a) Using Table 2, calculate approximate values of relative permeability gradient $\Delta B/\Delta H$ in the intervals 0.6 to 1.2 and 3.6 to 4.2 (both at 20°C). Explain what these values indicate about domain behaviour. [5]

(b) Determine the percentage reduction in B at $H = 3.0 \times 10^3$ A m^{-1} when temperature rises from 20°C to 60°C. State one implication for compass calibration. [3]

(c)

(i) Sketch one full cycle of drive current against time using the timing data for Variant 2. [3]

(ii) Deduce, with reasoning, the relative durations for which induced e.m.f. is positive, zero, and negative in one cycle for a single sensing coil. [4]

(d) Using Table 3, assess whether the device response remains linear up to 50 μT. Support your answer with numerical evidence using either gradient comparison or residuals from an estimated best-fit line. [5]

(e) In a simplified model,

$$V_{\text{out}} = K(T, f)\, B_{\text{ext}} \cos\theta,$$

where θ is the angle between sensor axis and external field, and K depends on temperature and drive frequency. Use the model to explain:

(i) why heading determination fails near certain orientations,

(ii) why multi-axis sensing reduces this failure.

[4]

(f) Open-ended evaluation: Propose a complete experimental method to separate the effects of *temperature drift*, *frequency response*, and *magnetic interference* on output accuracy. Your plan

should include controlled variables, measured quantities, uncertainty treatment, and success criteria for deployment. [6]

[Variant 2 Total: 30 marks]

Variant 2 – Suggested Answers (Marking Guidance)

(a) At 20°C: from $H = 0.6$ to 1.2, $\Delta B/\Delta H = (0.19 - 0.08)/(0.6 \times 10^3) \approx 1.83 \times 10^{-4}\,\mathrm{T\ m\ A^{-1}}$. From $H = 3.6$ to 4.2, $\Delta B/\Delta H = (0.54 - 0.53)/(0.6 \times 10^3) \approx 1.67 \times 10^{-5}\,\mathrm{T\ m\ A^{-1}}$. Large slope at low H indicates easy domain-wall motion; small slope at high H indicates near-saturation.

(b) At $H = 3.0 \times 10^3\,\mathrm{A\ m^{-1}}$, reduction is $(0.50 - 0.46)/0.50 \times 100\% = 8.0\%$. Calibration factor must be temperature-compensated or periodically re-zeroed.

(c) (i) Current waveform has rise, high plateau, fall, and zero plateau segments with the given durations. (ii) Since induced e.m.f. depends on $\mathrm{d}\Phi/\mathrm{d}t$, it is non-zero during rise and fall; near zero during both plateau segments. Polarity is opposite on rise and fall.

(d) Approximate low-field gradient using first three non-zero points is about $0.15\,\mathrm{mV}/\mu\mathrm{T}$, but high-field incremental gain from 40 to $50\,\mu\mathrm{T}$ is $(6.50 - 5.68)/10 = 0.082\,\mathrm{mV}/\mu\mathrm{T}$. This indicates compression/non-linearity near higher field.

(e) (i) When $\cos\theta \approx 0$ (near 90°), output tends to zero so angular resolution collapses and noise dominates. (ii) Orthogonal axes ensure at least one channel has substantial projection of field, allowing robust vector reconstruction of heading.

(f) Valid responses include factorial-style testing across temperature, frequency, and controlled interference source distance/orientation, with repeated measurements, error bars, and a deployment criterion such as RMS heading error below a stated threshold (e.g. $< 2°$).